

# Yupeng Su

Santa Barbara, CA, USA | [yupengsu@ucsb.edu](mailto:yupengsu@ucsb.edu) | [yupengsu.github.io](https://yupengsu.github.io) | [linkedin.com/in/yupeng-su](https://linkedin.com/in/yupeng-su)  
[github.com/YupengSu](https://github.com/YupengSu)

## Research Interests

---

**Efficient Pre-training / Post-training Optimization for Large Language Models:** focusing on quantization, sparsification, and low-rank decomposition techniques to improve training and inference efficiency.

**High-Performance Computing Infrastructure on GPU / FPGA Architectures:** including compiler optimization and GPU/FPGA kernel adaptation for high-performance, energy-efficient inference and practical edge deployment.

## Education

---

**University of California, Santa Barbara**, PhD in Computer Science – CA, USA Sept 2025 – present

- Advisor: Prof. Zheng Zhang (ECE, UCSB).
- GPA: 4.0/4.0.

**Southern University of Science and Technology**, BE in Microelectronics – China Sept 2021 – July 2025

- Advisor: Prof. Hao Yu (SME, SUSTech).
- GPA: 3.93/4.0, with 178 units completed.
- Awards: Nominee of Top Ten Graduates of SUSTech; Top Ten Graduates of the College of Engineering, SUSTech.
- Dissertation: *Enhanced Mix-Precision Post-Training Quantization for Large Language Models: Block Awareness and Outlier Identification*.

## Publications

---

**MuonQ: Enhancing Low-Bit Muon Quantization via Directional Fidelity Optimization** July 2026  
 Yupeng Su, Ruijie Zhang, Ziyue Liu, Yequan Zhao, Zheng Zhang  
[arxiv.org/abs/2605.11396](https://arxiv.org/abs/2605.11396) (CoLM, 2026)

**RankGuide: Tensor-Rank-Guided Routing and Steering for Efficient Reasoning** July 2026  
 Jiayi Tian, Yupeng Su, Ryan Solgi, Souvik Kundu, Zheng Zhang  
[arxiv.org/abs/2604.16694](https://arxiv.org/abs/2604.16694) (CoLM, 2026)

**Muon<sup>2</sup>: Boosting Muon via Adaptive Second-Moment Preconditioning** Apr 2026  
 Ziyue Liu, Ruijie Zhang, Zhengyang Wang, Yequan Zhao, Yupeng Su, Zi Yang, Zheng Zhang  
[arxiv.org/abs/2604.09967](https://arxiv.org/abs/2604.09967) (Under Review)

**MUON+: Towards More Effective Muon via One Additional Normalization Step for LLM Pre-training** Feb 2026  
 Ruijie Zhang, Yequan Zhao, Ziyue Liu, Zhengyang Wang, Yupeng Su, Liyan Tan, Zheng Zhang  
[arxiv.org/abs/2602.21545](https://arxiv.org/abs/2602.21545) (Under Review)

**TEON: Tensorized Orthonormalization Beyond Layer-Wise Muon for Large Language Model Pre-Training** Jan 2026  
 Ruijie Zhang, Yequan Zhao, Ziyue Liu, Zhengyang Wang, Dongyang Li, Yupeng Su, Sijia Liu, Zheng Zhang  
[arxiv.org/abs/2601.23261](https://arxiv.org/abs/2601.23261) (Under Review)

- APTQ+: Attention-FFN-aware Post-Training Quantization for a Layer-wise LLM Accelerator on FPGA** May 2026  
 Ziyi Guan, Zhengfei Chen, Dongkuan Wu, Ao Shen, Yifan Guo, **Yupeng Su**, Graziano Chesi, Mingqiang Huang, Ngai Wong, Hao Yu  
[10.1109/TCAD.2026.3697679](#) (IEEE TCAD, 2026)
- LLM-Barber: Block-Aware Rebuilder for Sparsity Mask in One-Shot for Large Language Models** Nov 2025  
**Yupeng Su**, Ziyi Guan, Xiaoqun Liu, Tianlai Jin, Dongkuan Wu, Zhengfei Chen, Graziano Chesi, Ngai Wong, Hao Yu  
[10.1109/ICCAD66269.2025.11240704](#) (IEEE/ACM ICCAD, 2025)
- PTQTP: Post-Training Quantization to Trit-Planes for Large Language Models** Sept 2025  
 He Xiao, Runming Yang, Qingyao Yang, Wendong Xu, Zhen Li, **Yupeng Su**, Zhengwu Liu, Hongxia Yang, Ngai Wong  
[arxiv.org/abs/2509.16989](#) (Under Review)
- Quantization Meets Reasoning: Exploring and Mitigating Degradation of Low-Bit LLMs in Mathematical Reasoning** Jan 2025  
 Zhen Li\*, **Yupeng Su**\*, Songmiao Wang, Runming Yang, Congkai Xie, Aofan Liu, Ming Li, Jiannong Cao, Yuan Xie, Ngai Wong, Hongxia Yang  
[arxiv.org/abs/2501.03035](#) (Under Review (\*Equal Contribution))
- EdgeLLM: A Highly Efficient CPU-FPGA Heterogeneous Edge Accelerator for Large Language Models** May 2025  
 Mingqiang Huang, Ao Shen, Kai Li, Haoxiang Peng, Boyu Li, **Yupeng Su**, Hao Yu  
[10.1109/TCSL.2025.3546256](#) (IEEE TCAS I: Regular Papers, 2025)
- APTQ: Attention-aware post-training mixed-precision quantization for large language models** June 2024  
 Ziyi Guan, Hantao Huang, **Yupeng Su**, Hong Huang, Ngai Wong, Hao Yu  
[10.1145/3649329.3658498](#) (IEEE/ACM DAC, 2024)

## Experience

---

- Student Research Assistant**, Prof. Hongxia Yang's Lab @ PolyU – Hong Kong, China Nov 2024 – May 2025
- Investigated the effects of compression on reasoning and explored data-driven strategies for restoration.
- Student Research Assistant**, Next Gen AI (NGai) Lab @ HKU – Hong Kong, China Aug 2024 – Feb 2025
- Contributed to lightweight networks for efficient inference under specified computing architectures.
- Summer Intern**, High Performance Integrated Circuit Design Lab @ SUSTech – China June 2024 – Sept 2024
- Implemented end-to-end pipelines for compressing, optimizing, and deploying models for efficient devices.
  - Published **EdgeLLM** based on the project and its results.